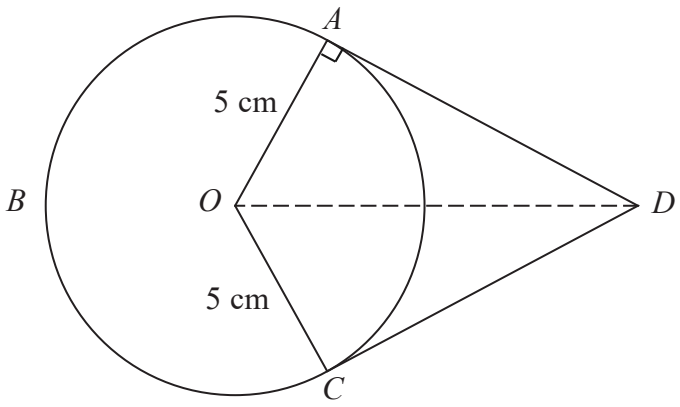
**18**




*A*, *B* and *C* are points on a circle of radius 5 cm, centre *O*.
*DA* and *DC* are tangents to the circle.
*DO* = 9 cm

Work out the length of arc *ABC*.
Give your answer correct to 3 significant figures.

Radius OA is perpendicular to tangent AD.

**1 mark**
Recognise angle OAD = 90°

Let $\theta$ = angle AOD. $\cos\theta = \frac{5}{9}$

**1 mark**
Adjacent ÷ hypotenuse

$\theta = \cos^{-1}(5 \div 9) = 56.251011...°$

**1 mark**
Use inverse cosine

Triangles OAD and OCD are congruent (RHS).

Minor angle AOC = $2\theta$; major angle = $360° - 2\theta$.

Arc ABC = $\frac{360 - 2\theta}{360} \times 2\pi \times 5$

**1 mark**
Use the major angle as a fraction of 360°

Arc ABC = 21.598272... cm ≈ 21.6 cm

**1 mark**
Round to 3 significant figures

21.6 cm

**(Total for Question 18 is 5 marks)**

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